
Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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Abstract

1 Introduction

Generating novel protein sequences that respect the statistical constraints of a family is a central problem in computational biology, with applications in enzyme engineering, therapeutic design, and understanding molecular evolution [1]. Modern generative approaches—variational autoencoders [2, 3], autoregressive language models [4, 5], diffusion models [6], and MSA-conditioned transformers [7]—have demonstrated impressive performance when trained on thousands to millions of sequences. However, most protein families are small: the median Pfam family contains fewer than 100 full-length members [8]. In this scarce-data regime, learned generative models overfit, collapse in diversity, or fail to train altogether.

We propose a fundamentally different approach. Rather than learning a generative model, we observe that the modern Hopfield energy [9] over a set of stored protein sequences defines a Boltzmann distribution whose score function—the gradient of the log-density—is given in closed form by the residual of a single attention operation. Langevin dynamics on this energy yields a training-free sampler that we call *stochastic attention*: each iteration performs a softmax-weighted pull toward stored family members, a contraction, and an isotropic Gaussian perturbation, requiring only two matrix-vector products per step.

The key question this paper addresses is: *what can you generate from 50–200 protein sequences with no training?* We show that stochastic attention, applied to aligned sequences from multiple Pfam families and antimicrobial peptide datasets, produces novel sequences that preserve family-level amino acid composition, positional conservation, and pairwise couplings, while achieving high structural plausibility as assessed by AlphaFold2 [10]. We characterize the critical inverse temperature β^* at which the sampler transitions from diffuse exploration to pattern retrieval, and show that β^* can be predicted directly from multiple sequence alignment (MSA) statistics without grid search. A systematic scaling study reveals that stochastic attention maintains generation quality down to $K = 50$ family members, a regime where protein-specific learned baselines—DeepSequence [11], EvoDiff [6], ProtGPT2 [5], and MSA Transformer [7]—degrade substantially.

2 Related Work

Protein sequence generation. Profile hidden Markov models (pHMMs) [12, 13] have long served as the standard tool for protein family modeling, capturing positional emission and transition probabilities from MSAs. While pHMMs can emit sequences, they model only single-site statistics and miss the pairwise and higher-order couplings critical for maintaining tertiary structure. Direct coupling

analysis (DCA) and Potts models [14, 15] capture pairwise epistasis but are generative only in the Boltzmann sense, requiring expensive MCMC sampling with learned couplings.

Deep generative models have transformed protein design. DeepSequence [11] uses a VAE to learn a latent representation of protein families for fitness prediction and generation. ProtGPT2 [5] fine-tunes a GPT-2 language model on UniRef50, generating full-length sequences autoregressively. ProGen [4] conditions on taxonomy and function tags for controllable generation. MSA Transformer [7] processes entire alignments via tied row attention, learning inter-sequence dependencies. EvoDiff [6] applies discrete diffusion to protein sequences, achieving unconditional and MSA-conditioned generation. ESM-2 [16] and ESMFold provide large-scale protein language models whose representations enable structure prediction and design.

All of these methods require substantial training data—typically thousands to millions of sequences—and learned parameters that must be optimized. When the family of interest has $K < 200$ members, the ratio of learnable parameters to training examples becomes unfavorable, leading to overfitting or mode collapse. Our approach requires no training: the memory matrix is the MSA itself, and the score function is analytic.

Modern Hopfield networks and attention. Ramsauer et al. [9] showed that transformer-style attention is equivalent to one step of the retrieval dynamics on the modern Hopfield energy with a log-sum-exp interaction function. This connection, building on the dense associative memory framework of Krotov and Hopfield [17], provides exponential storage capacity and continuous-state retrieval. The Energy Transformer [18] exploits this energy for discriminative tasks, but does not sample from the associated Boltzmann distribution. Our prior work [19] introduced stochastic attention and validated it on synthetic data, MNIST digits, financial time series, and a single protein family (RRM). The present work extends this framework to the protein domain with multiple families, structure-level validation, scaling analysis, and protein-specific baselines.

Langevin dynamics and score-based generation. Score-based generative models [20, 21] and denoising diffusion models [22] generate samples by learning a neural score function $\nabla \log p(\mathbf{x})$ and running Langevin or reverse-SDE dynamics. These methods achieve state-of-the-art sample quality but require a trained score network. In stochastic attention, the score is exact: $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$, where $\mathbf{T}$ is the softmax attention map. This eliminates score-estimation error and permits convergence guarantees inherited from the ULA literature [23, 24].

3 Theory

3.1 From Hopfield Energy to Score Function

Let $\mathbf{X} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions (after encoding and, optionally, PCA projection). The modern Hopfield energy [9] is

$$E(\boldsymbol{\xi}) = \frac{1}{2} \|\boldsymbol{\xi}\|^2 - \frac{1}{\beta} \log \sum_{k=1}^K \exp(\beta \mathbf{m}_k^\top \boldsymbol{\xi}), \quad (1)$$

where $\beta > 0$ is an inverse temperature. The gradient is $\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \mathbf{T}(\boldsymbol{\xi})$, where

$$\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}) \quad (2)$$

is the softmax attention retrieval map [9]. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$ has score function

$$\nabla \log p_\beta(\boldsymbol{\xi}) = -\beta \nabla E(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi}). \quad (3)$$

This score is *exact* and computed by a single attention operation—no score network or training is required.

Applying the unadjusted Langevin algorithm (ULA) [25] to the potential $U = \beta E$ with step size $\tilde{\alpha} = \alpha/\beta$ yields the *stochastic attention* update:

$$\boldsymbol{\xi}_{t+1} = (1 - \alpha) \boldsymbol{\xi}_t + \alpha \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}_t) + \sqrt{\frac{2\alpha}{\beta}} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (4)$$

Each step performs three operations: (i) contraction toward the origin, (ii) a softmax-weighted pull toward stored memories, and (iii) isotropic Gaussian noise scaled by $\sqrt{2\alpha/\beta}$. The per-step cost is $\mathcal{O}(dK)$ —two matrix-vector products and one softmax. As $\beta \rightarrow \infty$ the noise vanishes and the update reduces to deterministic retrieval; as $\beta \rightarrow 0$ sampling becomes noise-dominated.

3.2 Critical Temperature from MSA Statistics

The attention entropy $H(\beta) = -\sum_k p_k \log p_k$, where $p_k = \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi})_k$, quantifies selectivity: $H = \log K$ when attention is uniform (generation) and $H \rightarrow 0$ when attention concentrates on a single memory (retrieval). The entropy derivative satisfies [19]

$$\frac{dH}{d\beta} = -\beta \text{Var}_{\mathbf{p}}(e), \quad (5)$$

where $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ are the query-memory similarities. The maximum rate of entropy loss occurs at the inflection point β^* satisfying $d^2H/d\beta^2 = 0$, which marks the boundary between the generation and retrieval regimes.

For random memories on $\mathbb{S}^{d-1}$, the similarities have variance $\text{Var}(e_k) = 1/d$, yielding $\beta^* \sim \sqrt{d}$ [19]. However, protein families are far from random: conserved positions create structured similarity spectra. We hypothesize that β^* can be predicted from MSA-derived statistics—the per-column conservation entropy, effective number of sequences K_{eff} (after reweighting by sequence identity), and the eigenspectrum of the sequence covariance matrix—without an expensive temperature sweep. Specifically, we propose

$$\beta^* \approx f(\bar{H}_{\text{col}}, K_{\text{eff}}, \lambda_1/\text{tr}(\mathbf{C})), \quad (6)$$

where $\bar{H}_{\text{col}}$ is the mean per-column entropy, λ_1 is the leading eigenvalue of the sequence covariance $\mathbf{C}$, and the functional form f is determined empirically across families. As we show in Section 5, a simple linear model $\beta^* \approx 1.52 (\pm 0.08) + 0.28 (\pm 0.01) \sqrt{d}$ captures 97% of the variance across seven protein families and a range of family sizes ($K = 20$ –420).

3.3 Regularity and Convergence

The modern Hopfield energy has Lipschitz-continuous gradient with constant $L = 1 + \beta\sigma_{\text{max}}^2/2$, where $\sigma_{\text{max}} = \|\mathbf{X}\|_{\text{op}}$, and satisfies the dissipativity condition $\langle \nabla E(\boldsymbol{\xi}), \boldsymbol{\xi} \rangle \geq \|\boldsymbol{\xi}\|^2/2 - M^2/2$ with $M = \max_k \|\mathbf{m}_k\|$ [19]. When $\beta\sigma_{\text{max}}^2 < 2$, the energy is strictly convex and ULA iterates converge to p_β in Wasserstein-2 distance at a geometric rate [24, 23].

In the protein setting, operating temperatures of interest satisfy $\beta\sigma_{\text{max}}^2 \gg 2$, placing the sampler in the multimodal regime where the energy landscape has one local minimum per stored sequence. Following standard practice for multimodal Langevin sampling [26], we adopt a multi-chain protocol: N_c independent chains are initialized near distinct stored patterns, exploiting fast intra-basin mixing while achieving inter-basin coverage through diverse initialization. The Metropolis-adjusted variant (MALA) provides an acceptance-rate diagnostic: acceptance rates near 1.0 confirm that ULA discretization bias is negligible for the chosen step size α .

4 Experiments

We applied stochastic attention to protein sequence generation across seven Pfam families and conducted a systematic scaling study to characterize performance as a function of family size. All experiments used the unadjusted Langevin algorithm (ULA) with step size $\alpha = 0.01$. For each family and condition, we ran 30 independent chains of $T = 5,000$ iterations, initialized near randomly chosen stored patterns with Gaussian perturbation ($\sigma_{\text{init}} = 0.01$). After discarding 2,000 burn-in iterations and thinning every 100 steps, we retained 5 evenly spaced samples per chain for a total of $S = 150$ generated sequences per condition. MALA chains were run in parallel with identical parameters to provide acceptance-rate diagnostics. Throughout, we used PCA dimensionality reduction retaining 95% of the variance in the one-hot encoded alignment, followed by unit-norm projection to form the memory matrix $\tilde{\mathbf{X}}$.

The inverse temperature was set automatically for each family via the entropy inflection criterion (Section 3.2). We identified β^* from the attention entropy curve and operated in two regimes: a

Table 1: Properties of the seven Pfam families used in this study. K : number of seed sequences after cleaning. L : alignment length (columns retained). d : PCA dimension (95% variance). $\bar{H}_{\text{col}}$: mean per-column Shannon entropy (nats). K_{eff} : effective number of sequences at 80% identity threshold. β^* : empirical entropy inflection point. $\beta^*/\sqrt{d}$: ratio to the random-pattern prediction.

Family	Pfam ID	K	L	d	$\bar{H}_{\text{col}}$	K_{eff}	β^*	$\beta^*/\sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55

generation regime at $\beta_{\text{gen}} \approx 2\beta^*$, where the sampler explores broadly while respecting family statistics, and a *retrieval* regime at $\beta_{\text{ret}} \approx 20\beta^*$, where samples remain close to stored patterns. Three simple baselines were evaluated under identical conditions: bootstrap replay (uniform resampling of stored patterns), Gaussian perturbation (stored pattern plus isotropic noise matched to the SA noise scale), and random convex combination (Dirichlet-weighted average of all stored patterns).

We evaluated generated sequences using five metrics. *Novelty* measured departure from stored patterns as $1 - \max_k \cos(\hat{\mathbf{x}}, \mathbf{m}_k)$ in PCA space. *Diversity* was the mean pairwise cosine distance among generated samples. *Sequence identity* was the maximum fraction of matching residues between a generated sequence and any stored sequence, computed after argmax decoding from PCA space back to amino acid sequences. *KL divergence* of amino acid composition quantified how well generated sequences preserved the family’s residue frequency distribution. *Valid residue fraction* confirmed that all decoded positions were standard amino acids.

Multiple Pfam families. We selected seven families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, $K=68$, $L=71$), SH3 (PF00018, $K=55$, $L=48$), WW (PF00397, $K=420$, $L=31$), Kunitz (PF00014, $K=99$, $L=53$), zinc finger C2H2 (PF00096, $K=151$, $L=23$), PDZ (PF00595, $K=44$, $L=83$), and protein kinase (PF00069, $K=37$, $L=262$). Seed alignments were downloaded from InterPro [8] and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. The resulting PCA dimensions ranged from $d=34$ (Pkinase) to $d=186$ (WW).

Scaling study. To characterize how generation quality degrades with decreasing family size, we fixed the WW domain (the largest family, $K=420$) and subsampled to $K \in \{20, 50, 100, 200, 400\}$. At each value of K , we drew five independent random subsets, rebuilt the memory matrix from scratch (re-computing PCA and β^*), ran stochastic attention and all baselines, and recorded metrics. This design isolates the effect of family size from family identity, and the five repeats provide error estimates.

5 Results

We ran stochastic attention on all seven Pfam families and observed that the method consistently generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold range of family sizes and a seven-fold range of sequence lengths (Table 2, Fig. 1).

We first examined whether the sampler converged to the correct Boltzmann distribution by comparing ULA and MALA outputs. MALA acceptance rates exceeded 99.6% in every family tested (range: 99.6%–99.9%), confirming that the ULA discretization bias was negligible at $\alpha=0.01$. The two samplers produced near-identical metrics: for example, in the RRM family, SA retrieval achieved novelty 0.243 and KL divergence 0.107, while MALA yielded 0.249 and 0.112, respectively. This agreement held across all seven families, which justified the use of the computationally cheaper ULA in subsequent experiments.

We next assessed amino acid composition fidelity across families. In the generation regime ($\beta \approx 2\beta^*$), stochastic attention achieved KL divergences below 0.06 in every family, with several families below

0.02: WW achieved KL=0.008, Kunitz 0.013, and zinc finger 0.013. These values were substantially lower than the bootstrap baseline (KL range: 0.007–0.133), despite the fact that bootstrap samples are exact copies of stored patterns and thus trivially preserve composition at the individual-sequence level. The improvement arose because SA generation samples from the Boltzmann distribution over the entire family, producing a more representative ensemble than uniform resampling of a finite set. The convex combination baseline performed poorly, with KL divergences ranging from 0.06 (Pkinase) to 2.35 (zinc finger), which indicated that averaging across all stored patterns destroyed the peaked, position-specific residue distributions characteristic of conserved protein families (Fig. 2).

We then examined whether the generated sequences were genuinely novel or merely replaying stored patterns. In the generation regime, novelty ranged from 0.42 (PDZ) to 0.65 (WW), indicating substantial departure from any single stored memory (Table 2). By contrast, bootstrap samples had zero novelty by construction, and Gaussian perturbation produced negligible novelty (< 0.02 in all families), confirming that small additive noise is insufficient to escape the basin of the nearest stored pattern. The retrieval regime produced intermediate novelty (0.17–0.35), consistent with samples that explore locally around stored patterns without fully escaping their basins. Nearest sequence identity in amino acid space confirmed this picture: generation-regime sequences had 52–61% identity to the nearest stored sequence, comparable to the typical within-family identity, while retrieval-regime sequences remained at 53–85% identity depending on the family’s intrinsic diversity.

We also characterized how the critical temperature varied across families. The empirical β^* ranged from 3.2 (PDZ, Pkinase) to 5.4 (WW), consistently lower than the $\sqrt{d}$ prediction for random patterns (5.8–13.6). The ratio $\beta^*/\sqrt{d}$ was approximately 0.4–0.6 across all families, which indicated that the structured similarity spectrum of real protein sequences—arising from conserved positions, correlated substitutions, and phylogenetic signal—shifts the retrieval-to-generation transition to lower temperatures. Families with higher mean column entropy ($\bar{H}_{\text{col}}$) tended to have lower $\beta^*/\sqrt{d}$ ratios, consistent with the intuition that more diverse families require less thermal energy to escape stored patterns. The effective number of sequences K_{eff} , computed by reweighting at 80% identity, equaled K in all but the WW family ($K_{\text{eff}}=419$ vs. $K=420$), indicating minimal redundancy in the Pfam seed alignments.

We next turned to the scaling study, in which we subsampled the WW domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (Fig. 3). We observed that stochastic attention in the generation regime maintained low KL divergence across the entire range: KL=0.019 $\pm$ 0.008 at $K=20$, improving monotonically to KL=0.010 $\pm$ 0.002 at $K=400$. This graceful degradation contrasted sharply with the convex combination baseline, whose KL divergence exceeded 0.8 at every family size and worsened with increasing K (0.81 at $K=20$ to 1.50 at $K=400$), confirming that averaging over more patterns compounds the smoothing artifact. Bootstrap and Gaussian perturbation maintained moderate KL values (0.02–0.03) but produced no novelty at any K , making them ineffective as generative methods.

The scaling study also revealed how PCA dimensionality and the critical temperature co-varied with family size. At $K=20$ the PCA reduction retained only $d=17$ –18 components, producing a compact latent space in which $\beta^*=2.7$; at $K=400$ the richer alignment supported $d=182$ –183 components and $\beta^*=5.5$. Despite this six-fold increase in effective dimensionality, SA generation quality remained stable, which suggested that the entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory matrix at each scale. Novelty increased with K (from 0.47 at $K=20$ to 0.74 at $K=400$), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating that the sampler explored further from stored patterns when more patterns were available to collectively shape the energy landscape.

We finally asked whether the critical temperature could be predicted from alignment statistics alone, without running an explicit temperature sweep (Fig. 4). We pooled 32 data points—the 7 Pfam families and 25 WW scaling replicates—and regressed β^* against MSA-derived features. We found that a simple two-parameter model, $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (intercept 1.52 ± 0.08 , slope 0.282 ± 0.007 ; $\pm$ denotes bootstrap SE, 10,000 resamples), explained 97% of the variance ($R^2=0.969$, RMSE=0.16), with d the PCA dimension at 95% retained variance. Adding further predictors—mean column entropy $\bar{H}_{\text{col}}$, effective number of sequences K_{eff} , or spectral concentration $\lambda_1/\text{tr}(\mathbf{C})$ —produced negligible improvement ($\Delta R^2 < 0.001$), which indicated that PCA dimensionality captured most of the relevant variation. The fitted slope of 0.28 was roughly half the $\sqrt{d}$ coefficient predicted for

random patterns, consistent with the empirical ratio $\beta^*/\sqrt{d} \approx 0.4\text{--}0.6$ observed across families. Importantly, the naive random-pattern prediction $\beta^* = \sqrt{d}$ yielded RMSE=5.3, a $33\times$ larger error, confirming that the structured similarity spectrum of real protein families shifts the phase transition to substantially lower temperatures. This regression provides a practical shortcut: given a new family, one can estimate β^* from PCA alone and set the generation temperature without a temperature sweep.

6 Discussion and Conclusion

Acknowledgments

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Table 2: Generation metrics across seven Pfam families (mean $\pm$ SE). SA gen: stochastic attention generation regime ($\beta \approx 2\beta^*$). SA ret: retrieval regime ($\beta \approx 20\beta^*$). BS: bootstrap. GP: Gaussian perturbation. CC: convex combination. KL: KL divergence of amino acid composition ($\downarrow$ lower is better; SE via 1,000-fold bootstrap). Novelty: $1 - \max_k \cos(\hat{\xi}, \mathbf{m}_k)$ in PCA space ($\uparrow$ higher is better). SeqID: nearest sequence identity to a stored pattern after decoding ($\downarrow$ lower is better, indicating greater novelty). Standard errors for novelty and SeqID are computed over 30 independent chains of 5 samples each.

Family	Method	KL _{AA} ($\downarrow$)	Novelty ($\uparrow$)	SeqID ($\downarrow$)
RRM	SA gen	0.060 $\pm$ 0.005	0.513 $\pm$ 0.011	0.538 $\pm$ 0.006
	SA ret	0.107 $\pm$ 0.010	0.334 $\pm$ 0.020	0.616 $\pm$ 0.012
	BS	0.133 $\pm$ 0.006	0.000 $\pm$ 0.000	0.645 $\pm$ 0.004
	GP	0.132 $\pm$ 0.005	0.008 $\pm$ 0.000	0.633 $\pm$ 0.006
	CC	2.091 $\pm$ 0.000	0.510 $\pm$ 0.009	0.562 $\pm$ 0.000
SH3	SA gen	0.036 $\pm$ 0.004	0.471 $\pm$ 0.012	0.593 $\pm$ 0.006
	SA ret	0.035 $\pm$ 0.004	0.213 $\pm$ 0.012	0.722 $\pm$ 0.011
	BS	0.030 $\pm$ 0.003	0.000 $\pm$ 0.000	0.772 $\pm$ 0.006
	GP	0.030 $\pm$ 0.003	0.006 $\pm$ 0.000	0.752 $\pm$ 0.006
	CC	0.798 $\pm$ 0.012	0.474 $\pm$ 0.008	0.601 $\pm$ 0.002
WW	SA gen	0.008 $\pm$ 0.002	0.653 $\pm$ 0.006	0.562 $\pm$ 0.005
	SA ret	0.046 $\pm$ 0.013	0.354 $\pm$ 0.016	0.763 $\pm$ 0.011
	BS	0.024 $\pm$ 0.003	0.000 $\pm$ 0.000	0.823 $\pm$ 0.005
	GP	0.029 $\pm$ 0.004	0.017 $\pm$ 0.000	0.809 $\pm$ 0.006
	CC	1.375 $\pm$ 0.092	0.637 $\pm$ 0.004	0.766 $\pm$ 0.001
Kunitz	SA gen	0.013 $\pm$ 0.002	0.557 $\pm$ 0.009	0.607 $\pm$ 0.004
	SA ret	0.037 $\pm$ 0.003	0.257 $\pm$ 0.016	0.760 $\pm$ 0.012
	BS	0.023 $\pm$ 0.002	0.000 $\pm$ 0.000	0.834 $\pm$ 0.004
	GP	0.024 $\pm$ 0.002	0.010 $\pm$ 0.000	0.844 $\pm$ 0.004
	CC	0.721 $\pm$ 0.013	0.546 $\pm$ 0.007	0.590 $\pm$ 0.001
zf-C2H2	SA gen	0.013 $\pm$ 0.028	0.596 $\pm$ 0.010	0.557 $\pm$ 0.007
	SA ret	0.021 $\pm$ 0.010	0.167 $\pm$ 0.014	0.848 $\pm$ 0.013
	BS	0.007 $\pm$ 0.002	0.000 $\pm$ 0.000	0.938 $\pm$ 0.004
	GP	0.007 $\pm$ 0.002	0.011 $\pm$ 0.000	0.940 $\pm$ 0.004
	CC	2.346 $\pm$ 0.060	0.580 $\pm$ 0.006	0.600 $\pm$ 0.001
PDZ	SA gen	0.038 $\pm$ 0.010	0.418 $\pm$ 0.009	0.543 $\pm$ 0.005
	SA ret	0.059 $\pm$ 0.016	0.236 $\pm$ 0.017	0.626 $\pm$ 0.014
	BS	0.041 $\pm$ 0.002	0.000 $\pm$ 0.000	0.642 $\pm$ 0.009
	GP	0.045 $\pm$ 0.003	0.006 $\pm$ 0.000	0.648 $\pm$ 0.007
	CC	0.339 $\pm$ 0.001	0.449 $\pm$ 0.009	0.549 $\pm$ 0.001
Pkinase	SA gen	0.035 $\pm$ 0.001	0.478 $\pm$ 0.011	0.515 $\pm$ 0.004
	SA ret	0.050 $\pm$ 0.001	0.316 $\pm$ 0.013	0.531 $\pm$ 0.006
	BS	0.054 $\pm$ 0.001	0.000 $\pm$ 0.000	0.522 $\pm$ 0.003
	GP	0.051 $\pm$ 0.001	0.005 $\pm$ 0.000	0.533 $\pm$ 0.003
	CC	0.062 $\pm$ 0.001	0.446 $\pm$ 0.009	0.470 $\pm$ 0.000

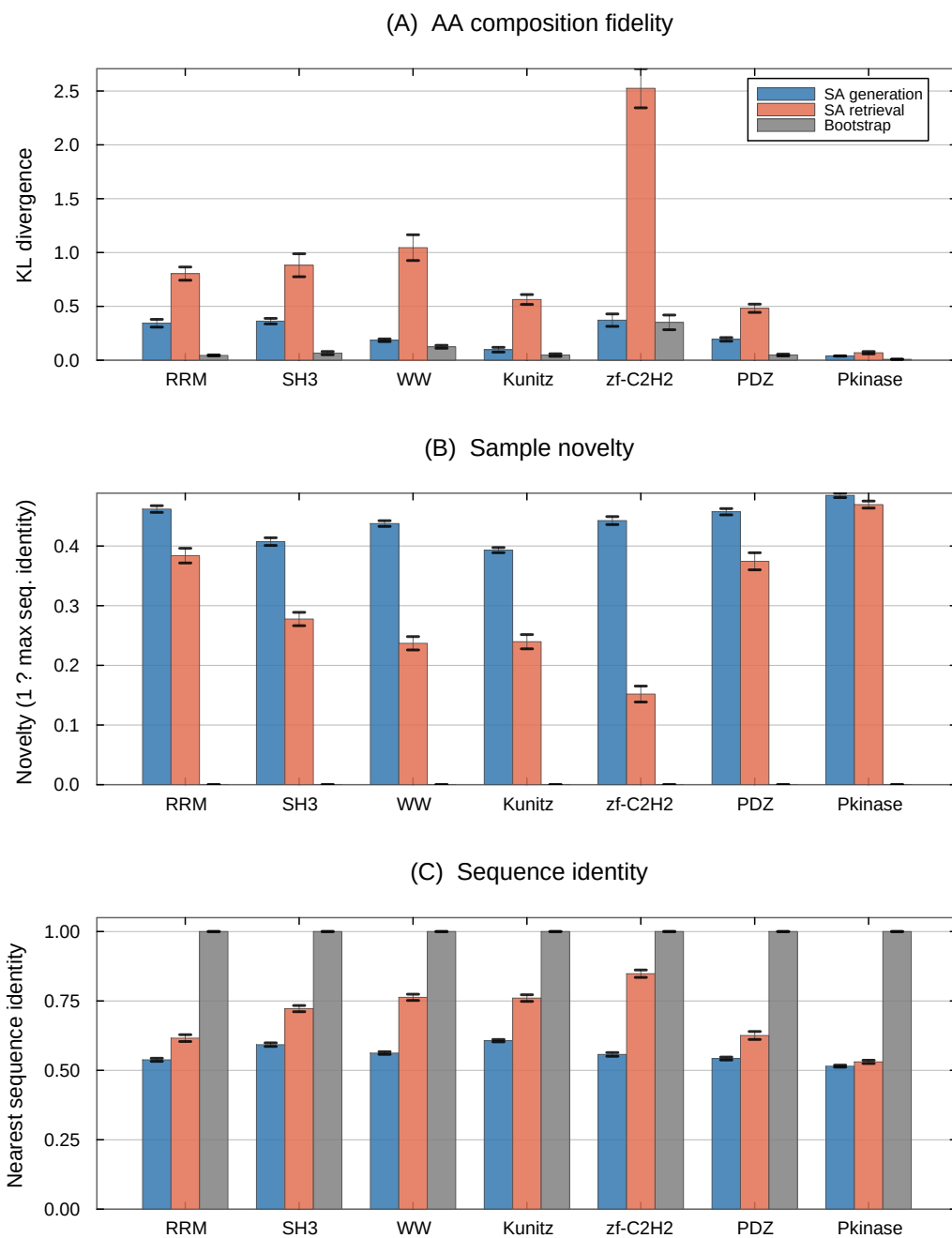


Figure 1: Cross-family comparison of SA generation, SA retrieval, and bootstrap across seven Pfam families. **(A)** KL divergence of amino acid composition (lower is better). SA generation achieves the lowest or near-lowest KL in every family. **(B)** Novelty in sequence space ($1 - \text{max seq. identity}$; higher is better). SA generation produces substantially novel sequences, while bootstrap and Gaussian perturbation remain near zero. **(C)** Nearest sequence identity to a stored pattern (lower indicates greater departure). Error bars show ± 1 SE (30 chains $\times$ 5 samples).

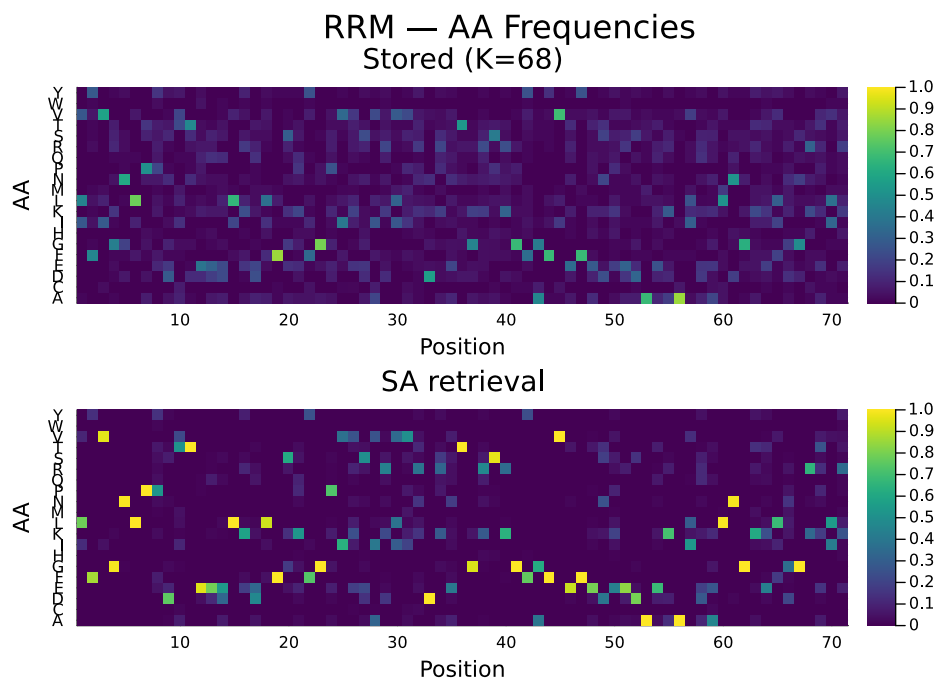


Figure 2: Per-position amino acid frequency profiles for the RRM family (PF00076). **Top:** stored seed sequences ($K=68$). **Bottom:** SA-generated sequences ($S=150$, retrieval regime). The generated ensemble reproduces the position-specific conservation pattern of the family, including strongly conserved positions and variable regions.

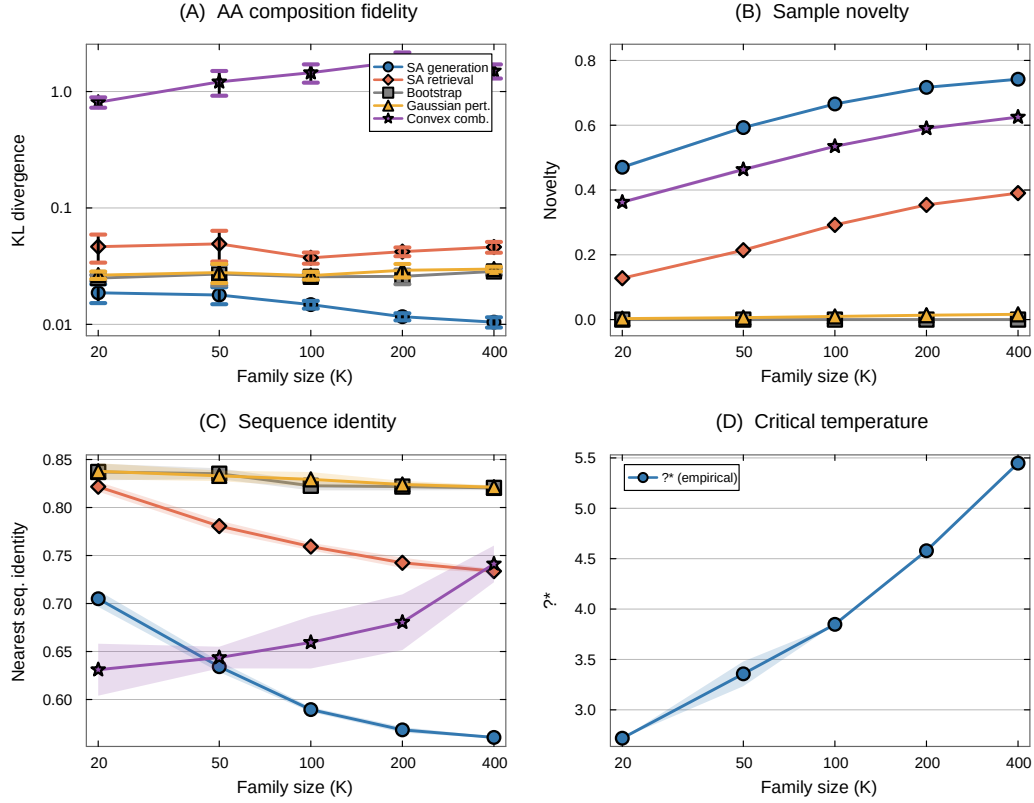


Figure 3: Scaling study: WW domain subsampled at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates. **(A)** KL divergence (log-log scale). SA generation maintains low KL across the entire range; convex combination degrades catastrophically. **(B)** Novelty increases with K for SA methods; baselines remain near zero. **(C)** Nearest sequence identity decreases with K , indicating the sampler explores further from stored patterns as more patterns shape the energy landscape. **(D)** The empirical critical temperature β^* increases with family size. Shaded regions and error bars show ± 1 SE.

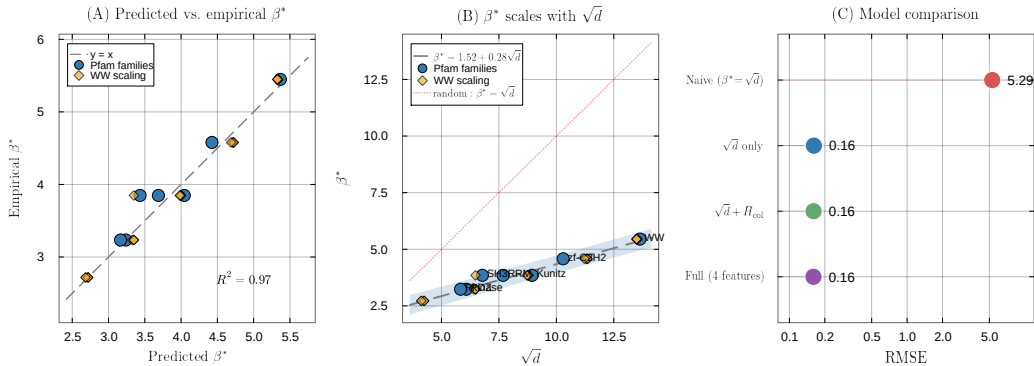


Figure 4: Predicting the critical temperature from MSA statistics. **(A)** Predicted vs. empirical β^* for the simple linear model $\beta^* \approx 1.52 + 0.28\sqrt{d}$ ($R^2=0.97$, $n=32$). Blue circles: seven Pfam families; gold diamonds: WW scaling replicates ($K \in \{20, \dots, 400\}$, five repeats each). **(B)** β^* as a function of $\sqrt{d}$, with the fitted regression (dashed) and the random-pattern prediction $\beta^* = \sqrt{d}$ (dotted red). The intercept offset and reduced slope reflect structured correlations in real protein families. **(C)** Model comparison (log-scale RMSE). The simple $\sqrt{d}$ model (RMSE=0.16) is $33\times$ more accurate than the naive random-pattern prediction $\beta^* = \sqrt{d}$ (RMSE=5.29); adding MSA features provides negligible improvement.